

NL Question Q

OFFLINE PRE-PROCESSING (zero API cost)

Phase 1: CHES PruningFAISS Top- k retrieval

$S' \subseteq S$

Phase 2: MCI-SQL EnrichmentMIN/MAX, cardinality, samples

C_{meta}

Phase 3: Context AssemblyDDL + MCI JSON + MAGIC guidelines

P_{enriched}

ASYMMETRIC LLM LOOP ($\sqrt{3}$ API calls)

Phase 4a: Generator_{llama-4-scout-17b} via Groq

y_0

Phase 4ab: ReflectorBack-translate SQL $\rightarrow$ English

Logic mismatch

OK

SemanticValidation

Syntax/Empty/Null err

Phase 4c: Resilient Critic_{gemini-2.5-flash}

PASS

y_{i+1}

Final SQL $\hat{y}$